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SUPERSEDING
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FEDERAL SPECIFICATION

STAND, LABORATORY APPARATUS SUPPORT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers laboratory apparatus support stands with metal or porcelain bases for buret supports and general laboratory use.

1.2 **Classification.**

1.2.1 *Types, grades, and sizes.* Laboratory apparatus stands shall be of the following types, grades, and sizes, as specified (see 6.2).

Type I—Rectangular-shaped base

Grade A—Metal
- Size 1—4 x 6 inches (in.)
- Size 2—5 x 8 in.
- Size 3—6 x 9 in.
- Size 4—6 x 11 in.

Grade B—Porcelain
- Size 1—9 x 9 in.
- Size 2—7 x 13 in.

Type II—Tripod-shaped base—Metal
- Size 1—3 1/4 in.
- Size 2—4 in.
- Size 3—5 in.
- Size 4—6 1/2 in.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-I-666—Iron Castings, Malleable.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-T-76—Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant, (For Carton Sealing).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services

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Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

National Bureau of Standards Handbook:

H-28—Screw Thread Standards for Federal Services.

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Metal bases.* The bases shall be made of ungalvanized malleable iron conforming to the requirements of grade I or II of QQ-I-666 or other suitable metal of sufficient weight to insure stability in use.

3.1.2 *Porcelain bases.* Porcelain bases shall be of high quality porcelain having an average coefficient of expansion between 20° and 200°C. of 0.0000036 centimeter (cm)/cm/1°C., gradually increasing to 0.0000047 cm/cm/1°C. at 1000°C. Softening point shall be at not less than 1400°C. and fusion point shall be not less than 1670°C. Porcelain shall be subjected to two firings to insure uniformity.

3.1.2.1 *Glaze.* Bases shall be glazed on top surface only. Glaze shall offer a high degree of chemical inertness and shall have a thermal coefficient of expansion approximately equal to that of the porcelain body of the tile.

3.1.3 *Rods.* Rods shall be made from steel, stainless steel commercially known as 18/8, or anodized aluminum.

3.2 Construction.

3.2.1 *Base.*

3.2.1.1 *Type I, grade A.* Type I, grade A bases shall be rectangular in design and conform to the dimensional requirements of table I. The four corners of the base shall be provided with legs which shall be cast integral with the base or securely attached to the base. The legs shall elevate the base approximately one inch. The top shall be flat and level and provided with a drilled hole through the midline of the base approximately 1 inch in from the edge of the short dimension. The hole shall be tapped with threads to fit the threaded end of the supporting rod and a heavy tapped boss should project approximately five-sixteenths of an inch below the under surface of the base of circumference to securely hold the supporting rod in place and to tend to equalize the strain over the entire surface of the base.

TABLE I. *Type I, grade A, rectangular-shaped base*

Size	Width, inch*	Length, inch*
1	4	6
2	5	8
3	6	9
4	6	11

Tolerances ± 1/8 inch.

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3.2.1.2 *Type I, grade B.* The type I, grade B base shall be rectangular in design and shall conform to the dimensional requirements of table II. The four corners of the base shall be provided with rubber feet securely fastened. The top shall be flat and level and provided with a drilled hole through the midline of the base approximately 1 inch in from the edge of the short dimension. The hole shall be of such dimension to accomodate the supporting rod.

TABLE II. *Type I, grade B, rectangular-shaped base*

Size	Width, inch*	Length, inch*
1	9	9
2	7	13

*Tolerance ± 1/8 inch.

3.2.2 *Type II.* The type II base shall be tripod in design and conform to the dimensional requirements of table III. The three legs shall radiate at angles of approximately 120 degrees. The legs shall taper from the point of contact to the center of the base to a degree which will raise the center of the base 1/2 ± 1/16 inch from the surface, and shall be provided with a drilled hole through the midline. The hole shall be tapped with threads to fit the threaded end of the supporting rod. On the under surface of the base directly beneath the drilled hole for the supporting rod, the casting shall be thickened and of sufficient circumference to securely and permanently hold the supporting rod firmly in place and tend to equalize the strain over the surface of the base.

TABLE III. *Type II, tripod-shaped base*

Size	Length of leg, inch*
1	3-1/4
2	4
3	5
4	6-1/2

*Tolerance ± 1/8 inch

3.3. **Supporting rods.** Rods for type I, grade A and type II shall conform to the dimensional requirements of table IV. Rods for type I, grade B shall conform to table V. Rods shall be threaded in accordance with Screw Thread Standards for Federal Services. Unified National Coarse Thread Series, Handbook H28. Threads of support rods shall be of sufficient length to permit insertion into complete depth of boss or of sufficient insertion into a partial boss in such a manner that the threaded portion of rod will extend sufficiently below partial boss to accomodate a nut, or of sufficient length to permit a secure attachment of rod to base by means of a furnished threaded collar and nut. The other end of rod shall be rounded.

TABLE IV. *Supporting rod, type I, grade A and type II*

Size	Diameter, inch*	Length, inch**	Threads per inch
1	5/16	18	18 UNC
2	3/8	20	16 UNC
3	1/2	24	13 UNC
4	1/2	36	13 UNC

*Tolerance ± 1/16 inch
**Tolerance ± 1/8 inch

TABLE V. *Supporting rod, type I, grade B*

Size	Diameter, inch*	Length, inch**	Threads per inch
1	1/2	23	13 UNC
2	1/2	23	13 UNC

*Tolerance ± 1/16 inch
**Tolerance ± 1/8 inch

3.4 **Assembly.** Each size supporting rod listed in table IV shall be suitable for use with each corresponding size base, type I, grade A and type II. Each size supporting rod listed in table V shall be suitable for use with each corresponding size base, type I, grade B. Base and supporting rods shall be furnished as a unit, as specified (see 6.2).

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3.5 **Finish.** Unless otherwise specified, the aluminum supporting rods shall be anodized, and the type I, grade A and type II bases finished in a black enamel which is chemical and corrosion resistant.

3.6 **Marking.** Each stand shall be permanently and legibly marked with the name or the trademark of the manufacturer.

3.7 **Workmanship.** Stands and supporting rods shall be free from defects (scratches, cracks, pits, crazing) or contamination (grease, oil) which mar their appearance or which might impair their use, durability, or serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 **Lotting.** A lot of laboratory apparatus support stands shall consist of those stands of the same type, grade, and size made by one manufacturer and submitted for delivery at one time.

4.4 **Sampling.**

4.4.1 *Nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.5 **Examination.** Sample support stands and level A and level B preparation for delivery requirements shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Support stands.*

Categories	*Defects*
Critical:	None defined.
Major:	*AQL 4.0 percent defective*
101	Dimensions incorrect.
102	Design incorrect.
103	Finish incorrect.
104	Defects (scratches, cracks, pits, crazing).
105	Contamination (grease, oil).
106	Marking incorrect, missing, or illegible.

4.5.1.2 *Level A and B preparation for delivery (section 5).*

Categories	*Defects*
Critical:	None defined.
Major:	*AQL 2.5 percent defective*
101	Unit or shipping container incorrect.
102	Cushioning or bracing incorrect.
103	Closure of unit or shipping containers incorrect.
104	Unlike stands packed per container.
105	Unit or shipping container damaged.
106	Marking of unit or shipping containers illegible, incorrect, or incomplete.

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5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Each stand disassembled shall be packaged in a box conforming to type I, class 1, of PPP-B-636. The box shall be provided with interior fittings or shall contain sufficient cushioning material to prevent shifting of or damage to contents. Box closures shall be secured with tape conforming to PPP-T-76.

5.1.2 *Level C.* Stands disassembled shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Stands shall be packed in shipping containers conforming to PPP-B-601, overseas type; or PPP-B-621, class 2. The gross weight of the shipping container shall not exceed 200 pounds.

5.2.2 *Level B.* Stands shall be packed in shipping containers conforming to PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class 2, W5c, or W5s, style RSC. Containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto. The gross weight of the shipping container shall not exceed the weight limitations of the applicable box specification or appendix thereto.

5.2.3 *Level C.* Stands shall be packed to insure carrier acceptable and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification rules and regulations of other carriers as applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** These laboratory apparatus stands are used for buret supports or general use in the laboratory.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Type, grade, and size of support stand required.
(d) Finish required on rods.
(e) Base and supporting rod combination required.

6.3 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard and military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 **Transportation description.** Transportation description applicable to these items is:

Stands, not otherwise indexed by name, other than furniture.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

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Stands, not otherwise indexed, other than furniture.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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